

NOVOSIBIRSK TOLMACHEVO, Russian Federation

WMO: 296340

Lat: 55.013N

Lon: 82.651E

Elev: 365

StdP: 14.50

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-33.1	-27.6	-40.3	0.6	-33.2	-34.4	0.8	-27.7	26.1	19.6	22.6	18.0	3.2	20	0.653

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	85.7	66.1	82.5	64.8	79.4	63.5	69.3	80.1	67.6	77.8	66.1	75.8	9.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.0	97.0	73.7	64.2	91.0	72.2	62.4	85.6	71.0	33.7	80.1	32.4	77.7	31.0	76.2	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.5	19.3	16.9	DB	-37.4	90.7	7.5	3.5	-42.8	93.3	-47.2	95.3	-51.5	97.3	-56.9	99.9	(4)
(5)			WB	-37.5	72.6	7.3	2.1	-42.8	74.1	-47.1	75.4	-51.2	76.6	-56.5	78.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	35.4	-0.3	5.9	19.9	38.8	53.0	63.0	66.3	61.9	50.6	37.6	19.5	6.9	(6)
(7)		DBStd	25.70	15.48	14.04	11.88	9.96	9.91	7.53	5.74	6.59	8.26	9.23	14.39	15.23	(7)
(8)		HDD50	6922	1559	1235	935	357	87	6	0	2	93	395	915	1338	(8)
(9)		HDD65	11020	2024	1655	1400	786	386	123	55	137	437	849	1365	1803	(9)
(10)		CDD50	1594	0	0	0	21	179	396	505	372	110	11	0	0	(10)
(11)		CDD65	218	0	0	0	0	13	63	95	42	5	0	0	0	(11)
(12)		CDH74	2361	0	0	0	5	256	703	856	463	78	0	0	0	(12)
(13)		CDH80	568	0	0	0	0	62	170	204	115	17	0	0	0	(13)
(14)	Wind	WSAvg	8.0	7.8	8.1	8.7	9.1	9.1	7.1	6.2	6.5	7.3	8.4	9.0	9.0	(14)
(15)	Precipitation	PrecAvg	18.5	0.9	0.7	0.8	1.0	1.4	2.2	2.8	2.3	1.6	1.7	1.6	1.4	(15)
(16)		PrecMax	25.9	1.9	1.5	1.7	1.9	2.9	4.8	4.4	5.7	3.6	3.2	2.7	2.4	(16)
(17)		PrecMin	12.8	0.2	0.1	0.2	0.3	0.1	0.7	0.4	0.4	0.6	0.6	0.9	0.6	(17)
(18)		PrecStd	3.3	0.4	0.4	0.4	0.5	0.8	1.1	1.2	1.1	0.7	0.7	0.6	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.8	35.4	45.5	73.3	86.4	88.0	89.4	88.0	82.3	68.3	48.0	35.5	(19)
(20)			MCWB	32.1	32.9	38.4	53.9	60.8	65.2	68.2	67.3	59.9	52.8	43.7	33.3	(20)
(21)		2%	DB	28.8	31.9	41.1	66.2	80.4	84.4	84.8	82.7	75.0	62.2	43.1	33.4	(21)
(22)			MCWB	27.4	29.8	35.7	50.7	58.3	64.8	67.3	65.9	58.2	49.9	39.7	32.1	(22)
(23)		5%	DB	25.8	28.7	37.8	60.7	76.2	81.2	82.1	79.0	69.9	55.7	39.5	30.2	(23)
(24)			MCWB	24.3	26.9	34.0	47.9	57.2	64.2	66.6	64.0	56.3	46.9	36.8	28.6	(24)
(25)		10%	DB	21.6	25.1	35.6	55.1	71.5	78.5	78.9	75.3	64.7	50.4	35.9	26.4	(25)
(26)			MCWB	20.0	23.6	32.4	45.4	55.2	63.2	65.3	62.4	53.9	44.7	33.8	24.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.1	32.9	39.4	55.8	64.0	70.0	72.3	70.3	63.0	53.9	44.9	33.3	(27)
(28)			MADB	33.4	34.3	44.3	67.0	81.4	81.5	82.1	82.3	77.0	64.7	46.9	34.2	(28)
(29)		2%	WB	27.8	29.8	36.6	51.9	61.0	67.9	70.1	67.6	59.6	50.9	39.9	31.6	(29)
(30)			MADB	29.0	31.4	40.0	62.7	75.1	79.9	80.3	79.3	70.9	59.3	42.4	32.7	(30)
(31)		5%	WB	24.2	26.9	34.5	48.6	58.5	66.1	68.4	65.8	57.5	48.1	36.8	28.4	(31)
(32)			MADB	25.7	28.5	37.5	60.1	71.9	77.1	77.9	75.8	68.2	55.0	39.2	30.0	(32)
(33)		10%	WB	20.5	23.9	32.4	45.4	56.0	64.4	67.1	63.8	55.1	45.2	33.9	24.8	(33)
(34)			MADB	21.9	25.5	35.3	54.2	69.1	74.8	76.1	72.6	63.2	50.4	35.7	26.2	(34)
(35)	Mean Daily Temperature Range	MDBR	14.5	16.1	17.1	18.2	22.0	20.4	19.5	20.3	19.1	14.9	12.0	13.4	(35)	
(36)		5% DB	MADB	14.2	14.5	16.3	27.2	28.9	26.3	24.2	26.4	27.3	24.5	11.9	12.7	(36)
(37)			MCWBR	14.0	13.7	12.6	15.0	13.0	11.4	10.5	11.8	13.2	14.3	9.6	12.7	(37)
(38)		5% WB	MADB	14.2	13.8	13.9	25.0	25.3	22.4	20.5	22.8	24.6	21.8	11.7	12.9	(38)
(39)			MCWBR	14.0	13.5	11.4	15.0	13.3	11.0	10.2	11.5	12.6	13.8	9.9	13.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.294	0.328	0.378	0.410	0.407	0.410	0.439	0.413	0.373	0.355	0.327	0.297	(40)	
(41)		taud	2.005	1.960	1.912	2.085	2.213	2.288	2.207	2.261	2.320	2.235	2.046	1.954	(41)	
(42)		Ebn at Noon	186	224	241	255	265	265	254	253	248	218	170	150	(42)	
(43)		Edh at Noon	23	34	46	44	42	39	42	37	31	26	22	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	297	625	1092	1473	1693	1906	1708	1439	970	488	238	190	(44)	
(45)		RadStd	26	36	79	94	184	172	112	108	115	64	42	30	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page